Primitive Operation Descent in Phase Calculus

From Survivor Mark to Arithmetic, π, Calculus, and EML

A Preprint

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April 22, 2026

Abstract

This note gives the missing bridge for the EML placement paper. EML begins with a continuous node $\exp(x) - \ln(y)$. Phase Calculus begins lower. The operation layer descends through the carried state:

$$\begin{aligned}&\text{survivor mark} \to \text{primitive roll} \to i, \theta, \pi\\ &\to \mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C}\\ &\to q, c \to \{Q, B, L\} \to \Pi \circ E = G \circ \Pi\\ &\to \text{continuous shadow calculus} \to \text{eml}.\end{aligned}$$

The note therefore proves the exact placement: EML is a continuous-shadow composite, while Phase Calculus is the discrete lifted-state grammar underneath the expression layer. In the restored main paper, the native operator is the selector evolution itself; this bridge note isolates the descent spine that that operator and its Red quotient share.

1 Machine-side survivor witness

```
0F BD C0  ; BSR EAX, EAX
F7 D0     ; Bit-flip / roll
0F BD C0  ; BSR EAX, EAX
F7 D0     ; Bit-flip / roll

0F BD C0    ; failed search on zero -> ZF=1
0F 94 05 00 ; SETZ writes 1 when ZF=1
```

Lemma 1 (Survivor mark)**.**

$$\mathrm{BSR}(0) \Rightarrow \mathrm{ZF} = 1, \qquad \mathrm{SETZ}(1) = 1.$$

Proof. The search instruction has no set-bit target on zero and sets the zero flag. The write instruction SETZ emits 1 exactly when that flag is set. □

This is an operational witness. The theorem burden comes from the formal stack: failed same-class closure does not become terminal nullity; it produces the survivor needed for lawful continuation.

2 Primitive roll and native phase constants

Definition 1 (Primitive roll)**.**

$$z(0) = i, \qquad \frac{dz}{d\theta} = iz, \qquad z(\theta) = ie^{i\theta}.$$

The visible witnesses are

$$z(0) = i, \quad z\left(\frac{\pi}{2}\right) = -1, \quad z(\pi) = -i, \quad z\left(\frac{3\pi}{2}\right) = 1, \quad z(2\pi) = i.$$

Definition 2 (Half-turn constant)**.**

$$\pi := \inf\{\theta > 0 : R_\theta = -I\}.$$

Theorem 1 (Native phase layer)**.** *The objects i, θ, and π are native to the Phase Calculus descent before EML appears.*

Proof. The roll begins at i, so i is the source witness. The equation $z' = iz$ uses θ as its continuation parameter. The half-turn constant is defined by the first positive opposite-pole attainment of the continuous ORS completion. None of these is introduced by the EML node. □

3 Arithmetic operation descent

The primitive-origin stack licenses algebra in stages:

$$0D \to 1D \to 2D,$$

then the carrier-stage completion gives

$$\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R}, \qquad \mathbb{C} = \mathbb{R} \oplus i\mathbb{R}.$$

Object/operation	Descent source	Consequence
1	survivor mark / proto-$\mathbb{N}$	seed constant below continuous syntax
Addition	successor and serial articulation	additive operation layer
Subtraction	group completion $\mathbb{N} \to \mathbb{Z}$	$a - b$ available before EML
Multiplication	ring/field completion	product and powers downstream
Division	$\mathbb{Q}$ and $\mathbb{R}$ completion	ratios and real branch coordinates
i	ORS quarter-turn $R^2 = -1$	complex branch native
π	least positive half-turn	half-turn native

Theorem 2 (Arithmetic operation descent)**.** *The arithmetic operations required by EML are not primitive inputs in Phase Calculus. They descend from proto-$\mathbb{N}$, group completion, field completion, and the ORS complex branch.*

Proof. Proto-$\mathbb{N}$ supplies the serial successor structure. Group completion supplies additive inverses and subtraction. Ring and field completion supply multiplication, division, and real arithmetic. The two-axis ORS algebra supplies $\mathbb{C} = \mathbb{R} \oplus i\mathbb{R}$. Therefore the arithmetic operations used by EML are already downstream of the primitive-origin stack. □

4 Carried finite-resolution memory

Definition 3 (Lifted state)**.**

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \qquad q = (u, v), \qquad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv}\right].$$

The exact carried remainder is

$$r = \frac{1}{uv}.$$

Balanced refinement is

$$B(u, v) = \mathrm{sort}(v, u + v),$$

and on the Fibonacci corridor,

$$B^n(1, 1) = (F_{n+1}, F_{n+2}), \qquad r_n = \frac{1}{F_{n+1}F_{n+2}}.$$

Theorem 3 (Carried remainder memory)**.** *The pair $q = (u, v)$ and germ c carry finite-resolution arithmetic information that is invisible to a scalar expression value.*

Proof. The visible phase is $ie^{i\theta}$. The carried germ depends additionally on uv. Two lifted states may share θ while differing in (u, v), and therefore differing in unresolved width $2\pi/(uv)$. Hence (q, c) are state memory, not expression syntax. □

5 Quotient descent to continuous calculus

Definition 4 (Exact quotient criterion)**.**

$$\Pi \circ E = G \circ \Pi.$$

Here E is lifted-state evolution, Π is a branch projector, and G is the branch-visible law.

On the continuous shadow branch, ordinary calculus is recovered as branch language. The operations relevant to EML are then:

$$\mathrm{Sub}_{\mathrm{PC}}(a,b) = a - b,$$
$$\mathrm{Exp}_{\mathrm{PC}}(x) = e^x, \qquad \mathrm{Log}_{\mathrm{PC}}(y) = \ln y.$$

Dependency-wise, $\mathrm{Exp}_{\mathrm{PC}}$ is the shadow flow solving $f' = f$, $f(0) = 1$, while $\mathrm{Log}_{\mathrm{PC}}$ is its positive-branch inverse.

Theorem 4 (Analytic operation descent)**.** *$\mathrm{Sub}_{\mathrm{PC}}$, $\mathrm{Exp}_{\mathrm{PC}}$, and $\mathrm{Log}_{\mathrm{PC}}$ are continuous-shadow branch operations. They are not primitive objects of the lifted-state grammar.*

Proof. Subtraction belongs to the descended real additive group. The continuous shadow theorem supplies ordinary differential calculus on the selected branch. The exponential is the branch solution of $f' = f$, $f(0) = 1$, and logarithm is the inverse branch. Therefore all three operations are branch descendants. □

6 Primitive operation descent theorem

Theorem 5 (Primitive operation descent)**.** *The operation layer relevant to EML descends in Phase Calculus as*

$$\begin{aligned}
&\textit{survivor mark} \to \textit{primitive roll} \to i, \theta, \pi \\
&\to \mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C} \\
&\to q = (u,v),\ c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv}\right] \\
&\to \{Q, B, L\} \to \Pi \circ E = G \circ \Pi \\
&\to \mathrm{Sub}_{\mathrm{PC}}, \mathrm{Exp}_{\mathrm{PC}}, \mathrm{Log}_{\mathrm{PC}} \to \mathrm{eml}_{\mathrm{PC}}\,.
\end{aligned}$$

Consequently,

$$\mathrm{eml}_{\mathrm{PC}}(x,y) = \mathrm{Sub}_{\mathrm{PC}}(\mathrm{Exp}_{\mathrm{PC}}(x), \mathrm{Log}_{\mathrm{PC}}(y)) = e^x - \ln y.$$

Proof. The previous sections prove each dependency arrow. The survivor witness gives the first mark. The roll gives native phase constants. The completion chain gives arithmetic and complex operations. The lifted object carries finite-resolution state. The operator core evolves that state. Quotient descent selects the continuous branch. The continuous branch supplies subtraction, exponential, and logarithm. Their composite is exactly the EML node. □

7 Native π operational surface

The native π package separates three modes:

streaming prefix, bank-indexed random access, no-bank scalar interval decoding.

The locked evidence records

1,000,000 native decimal digits in 1.65086778 s, $D_{\mathrm{safe}} = 1{,}366{,}163$,

100,000 digits in 0.08884997 s, 1,125,492.76 digits/s,

and, for certified bank-indexed 1024-digit blocks,

$$5.3941900 \times 10^{-7} \text{ s/query}, \qquad 1.8983388 \times 10^{9} \text{ digits/s}.$$

Theorem 6 (Certified bank access)**.** *If a native bank has length D and certificate $D_{\mathrm{safe}} \geq D$, then any query (n, ℓ) with*

$$n + \ell - 1 \leq D$$

returns only certified digits.

Proof. Since $D \leq D_{\text{safe}}$, every digit inside the bank is inside the certified stable prefix. The query condition keeps the requested block inside the bank. $\square$

The no-bank scalar-collapse obstruction is instead

$$2b^{n-1}B_m < b^{-\ell}, \qquad B_m < \frac{1}{2}b^{-(n+\ell-1)}.$$

For decimal precision $B_m = 10^{-D_m}$, this gives

$$D_m > (n+\ell-1)\log_{10} b + \log_{10} 2.$$

This is a linear warm-up obstruction for scalar interval decoders, not a limit on branch-retaining lifted-state access.

8 Conclusion

EML compresses continuous calculator syntax. Phase Calculus derives the carried-state operation layer beneath that syntax:

$$\boxed{\text{EML is the continuous expression node.}}$$

$$\boxed{\text{Phase Calculus is the primitive lifted-state descent beneath it.}}$$